



Jirapat Chaumaungphan

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Portfolio

SUMMARY

Robotics Engineering student with a strong passion for ROS2, Computer Vision, and Autonomous Systems. Proven logical problem-solving skills through the development of a 5-DOF robotic arm (WhitePost). Experienced in Python/C++

EDUCATION

Bachelor of Engineering (B.Eng.) in Robotics Engineering and AI

Jun 2022 - Mar 2026

Relevant Coursework : Robot Design Essential, Computer Vision, Robotic Engineering, System Analysis and Control, Embedded Systems, ROS2 , AI for Robotics

Chiang Mai University
GPAX : 3.05

EXPERIENCE & INTERNSHIP

Jinpao Automation Contest 2023

Nov 2022 - Feb 2023

- Designed the robot's mechanical chassis and structural components for optimal stability.
- Collaborated with a multidisciplinary team to implement an automated system for researcher tasks.

IEEE Region 10 Robotics Competition 2024

Mar 2024 - May 2024

- Qualified as Stage 2 Finalist in a international robotics competition.
- Developed an autonomous Floating Robot (ASV) for collect environmental water quality sampling.

TESR Co.,Ltd. Internship

Apr 2025 - May 2025

- Developed a high-payload Autonomous Mobile Robot (AMR) capable of transporting up to 300 kg.
- Collaborated on hardware-software integration by interfacing ROS2 with ESP32 via Serial communication to control Hub Servo Motor drivers.

PROJECT

Turtlebot3 : SLAM Navigation

Jan 2024 - Feb 2024

- Deployed SLAM (Lidar-based) and Nav2 stack for autonomous environment mapping and path planning
- Visualized and debugged transform trees (TF) and laser scan data via Rviz2 and Gazebo.

Orange Robot : a customized mobile robot platform

Feb 2024 - Mar 2024

- Developed a mobile robot platform integrated with a 5-DOF robotic arm for a Teleoperation system.
- Simulated robot kinematics and navigation in Gazebo using ROS2 .
- Manually navigated the robot to target locations and operated the robotic arm to complete precise orange box pick-up and drop-off tasks.

Service Robot for Dental Clinic CMU

Jun 2024 - Oct 2024

- Designed and developed a Service Robot prototype for dental clinic environments, integrated with a 5-DOF robotic arm.
- Developed a prototype mobile robot with an arm to manually transport dental trays from the sterilization room to treatment rooms.

White Post : a robot document delivery

Jun 2025 - Mar 2026

- Designed the custom robot chassis to support a 5-DOF arm and integrated a tracked and wheel drive system for multi-terrain mobility.
- Developed full control software for a 5-DOF robotic arm, implementing Inverse Kinematics (IK) and Forward Kinematics (FK) for precise button lift handling.
- Developed and deployed a custom YOLOv8 model integrated with OpenCV to identify and track elevator buttons and monitor elevator door status (Open/Closed).

SKILLS

- Programming Language :** Python, C++, C, Lua
- Robotics Technical Skills :**
 - Hardware:** TurtleBot3 Burger/Waffle, 2D LiDAR, OpenCR Controller, Raspberry Pi 5/Zero 2W, ESP32, Arduino Uno
 - Navigation & SLAM:** SLAM (Lidar-based), Nav2, Path Planning (RRT, RRT*, A*)
 - Frameworks & Tools:** ROS2 (Foxy, Humble), Rviz2, Gazebo Classic
- Computer Vision & AI:** OpenCV, YOLOv8, Image Processing, Roboflow, Large Language Models (ChatGPT/Gemini API Integration),
- Industrial Automation:** Siemens PLC , HMI Design, TIA Portal v16, Ladder Logic
- Language:** Thai (Native) , English (Intermediate EF Set 45/100 Reading,Listening)
- Tools & Development:** Git/GitHub, Linux/Ubuntu, Visual Studio Code, SolidWorks, EasyEDA, Arduino IDE
- Soft Skills:** Logical Problem Solving , Technical Communication, Team Collaboration, Rapid Self-Learning

JIRAPAT CHAUMAUNGPHAN

has successfully completed the EF SET Certificate and has earned level:



✓ Reading ✓ Listening

Awarded on: 15 Dec 2025

Understanding the results

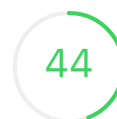
EF SET	0-20	21-30	31-40	41-50	51-60	61-70	71-100
CEFR	A0 Novice	A1 Beginner	A2 Elementary	B1 Intermediate	B2 Upper Intermediate	C1 Advanced	C2 Proficient

The achieved level is 45/100 on the EF SET score scale and B1 Intermediate according to the Common European Framework of Reference (CEFR). The EF SET score is calculated as an average of the skill section scores.



Reading

B1 Intermediate



Listening

B1 Intermediate